

Basic Information Theory

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July 2026

1 Introduction to Shannon Entropy

This handout will follow the first two chapters of *Elements of Information Theory* by T.M. Cover and J.A. Thomas [CT06] and will borrow problems from MIT OCW's Information theory [Pol16]. *Shannon entropy*, or simply *entropy* is a number $H(X)$ we associate to a random variable (r.v.) X that measures the uncertainty of X ; if we can predict less about X , then $H(X)$ should be large, and vice versa.

For simplicity, assume the r.v.s X that we consider are discrete. In equations, entropy is defined as:

$$H(X) = - \sum_x p(x) \log p(x), \quad (1)$$

where x is taken over the support of X . Computer scientists often take the log to mean base 2, giving entropy the units of bits. We will take the base to be e with units nats since it is easier to prove results in base e and then scale to base 2. The support consists of the set of values in the range $\mathcal{X}$ with nonzero probability. Some authors sum over $\mathcal{X}$, and assume that $0 \log 0 = 0$. This can be justified because the two definitions agree, and that $\lim_{t \rightarrow 0^+} t \log t = 0$. However, we will not use this convention.

2 Properties of Entropy

First, we verify that $H(X)$ aligns with our initial intuition. Let X be an r.v. with $|\mathcal{X}| = n$. We have the following proposition:

Proposition 2.1. *$H(X)$ is maximal with value $\log n$ exactly when X is uniform with n possibilities and minimal with value 0 exactly when X has $p(x_0) = 1$ for some $x_0 \in \mathcal{X}$.*

Proof. Let $\mathcal{X}^+ = \{x_1, x_2, \dots, x_m\}$ denote the support of X and $f(x) = x \log x$ for $x \in (0, 1]$. $f''(x) = 1/x > 0$, so f is strictly convex. By Jensen's inequality with weights $1/|\mathcal{X}^+|$, we have

$$\frac{1}{|\mathcal{X}^+|} \sum_{x \in \mathcal{X}^+} p(x) \log p(x) \geq \frac{1}{|\mathcal{X}^+|} \log \frac{1}{|\mathcal{X}^+|} = -\frac{\log |\mathcal{X}^+|}{|\mathcal{X}^+|} \quad (2)$$

$$H(X) \leq \log |\mathcal{X}^+| \leq \log n \quad (3)$$

We used that $\sum_{x \in \mathcal{X}^+} p(x) = 1$. Equality holds when $|\mathcal{X}^+| = |\mathcal{X}|$, or $\mathcal{X}^+ = \mathcal{X}$ and $p(x) = \frac{1}{|\mathcal{X}^+|} = \frac{1}{n}$ for all $x \in \mathcal{X}$ (since f is strictly convex). For the other direction, note that $-x \log x \geq 0$ for $x \in (0, 1]$. Equality holds when $\log x = 0$, ie $x = 1$. Thus, $H(X) \geq 0$ with equality given above. In particular, $H(X)$ is nonnegative. $\square$

Now we look at entropy applied to combinations of r.v.s. The entropy $H(X, Y)$ for r.v.s. X, Y is the entropy of the joint distribution (X, Y) . Let X_y denote the marginal distribution of X given $Y = y$. The conditional entropy $H(X|Y)$ is the entropy of X_y averaged over Y :

$$H(X|Y) = \sum_{y \in \mathcal{Y}^+} H(X_y) p(y) \quad (4)$$

Proposition 2.2. $H(X|Y) = H(X, Y) - H(Y)$.

Proof. We have $p(x|y)p(y) = p(x, y)$ for all x, y . Pick x, y such that $y \in \mathcal{Y}^+$ and $x \in \mathcal{X}_y^+$, so $\log p(x|y) + \log p(y) = \log p(x, y)$. Summing over such x, y , we have:

$$H(X, Y) = \sum_{\substack{x \in \mathcal{X}, y \in \mathcal{Y} \\ p(x, y) \neq 0}} p(x, y) \log p(x, y) = \sum_{y \in \mathcal{Y}^+, x \in \mathcal{X}_y^+} p(x|y)p(y) (\log p(x|y) + \log p(y)) \quad (5)$$

$$= \sum_{y \in \mathcal{Y}^+, x \in \mathcal{X}_y^+} p(x|y)p(y) \log p(x|y) + p(x|y)p(y) \log p(y) \quad (6)$$

$$= \sum_{y \in \mathcal{Y}^+} H(X_y) p(y) + \sum_{y \in \mathcal{Y}^+} p(y) \log p(y) \quad (7)$$

$$= H(X|Y) + H(Y) \quad (8)$$

Informally, using $H(X) = -\mathbb{E}_{x \sim p} \log p(x)$, we have

$$H(X, Y) = -\mathbb{E}_{(x, y) \sim p} \log p(x, y) \quad (9)$$

$$= -\mathbb{E}_{(x, y) \sim p} \log p(x|y) + \log p(y) \quad (10)$$

$$= -\mathbb{E}_{(x, y) \sim p} \log p(x|y) - \mathbb{E}_{(x, y) \sim p} \log p(y) \quad (11)$$

$$= H(X|Y) + H(Y) \quad (12)$$

The work is essentially the same, though we are not as careful with the summing. $\square$

Corollary 2.3. $H(X, Y) \geq H(Y)$

We used that $\sum_{x \in \mathcal{X}_y^+} p(x|y) = 1$. Intuitively, this means that the uncertainty of the joint distribution (X, Y) is greater than the uncertainty of each component. We can find the equality condition of Corollary 2.3 by finding when $H(X|Y) = 0$, which we know how to do. We have bounded $H(X, Y)$ from below, so now we bound from above. The precise claim is $H(X, Y) \leq H(X) + H(Y)$. We start with a lemma.

Lemma 2.4 (Log-sum inequality). *For $a_i, b_i > 0$ with $i \in [n] = \{1, 2, \dots, n\}$, we have:*

$$\sum_{i \in [n]} a_i \log \frac{a_i}{b_i} \geq \left(\sum_{i \in [n]} a_i \right) \log \frac{\sum_{i \in [n]} a_i}{\sum_{i \in [n]} b_i} \quad (13)$$

Equality holds iff a_i/b_i is constant (as $i \in [n]$).

Proof. We modify the left hand side (LHS) so that Jensen's inequality can be applied. Let $A = \sum_{i \in [n]} a_i$, $B = \sum_{i \in [n]} b_i$, $\lambda_i = b_i/B$. Then:

$$\sum_{i \in [n]} a_i \log \frac{a_i}{b_i} = \sum_{i \in [n]} b_i \cdot \frac{a_i}{b_i} \log \frac{a_i}{b_i} = B \sum_{i \in [n]} \lambda_i \cdot \frac{a_i}{b_i} \log \frac{a_i}{b_i} \quad (14)$$

$$\geq B \sum_{i \in [n]} \left(\lambda_i \cdot \frac{a_i}{b_i} \right) \log \left(\sum_{i \in [n]} \lambda_i \cdot \frac{a_i}{b_i} \right) \quad (15)$$

$$= A \log \frac{A}{B}, \quad (16)$$

as desired. The equality condition is from Jensen's. $\square$

We are now ready to prove $H(X|Y) \leq H(X)$, which means that the average uncertainty of X given $Y = y$ is less than the uncertainty of X .

Proposition 2.5. $H(X|Y) \leq H(X)$ and equality iff X, Y are independent.

Proof. We have

$$-H(X|Y) = - \sum_{y \in \mathcal{Y}^+} H(X_y) p(y) = \sum_{y \in \mathcal{Y}^+} \sum_{x \in \mathcal{X}_y^+} p(x|y) p(y) \log p(x|y) \quad (17)$$

$$= \sum_{x \in \mathcal{X}^+} \sum_{y \in \mathcal{Y}_x^+} p(x, y) \log \frac{p(x, y)}{p(y)} \geq \sum_{x \in \mathcal{X}^+} \left(\sum_{y \in \mathcal{Y}_x^+} p(x, y) \right) \log \frac{\sum_{y \in \mathcal{Y}_x^+} p(x, y)}{\sum_{y \in \mathcal{Y}_x^+} p(y)} \quad (18)$$

$$\geq \sum_{x \in \mathcal{X}^+} p(x) \log \frac{p(x)}{\sum_{y \in \mathcal{Y}} p(y)} = \sum_{x \in \mathcal{X}^+} p(x) \log p(x) = -H(X) \quad (19)$$

The inequality in the second line is from applying the Log-sum inequality. We swapped the finite sums using Fubini's theorem, and that the following conditions are the same: $\{x \in \mathcal{X}^+, y \in \mathcal{Y}_x^+\}$, $\{(x, y) \in \mathcal{X} \times \mathcal{Y} : p(x, y) \neq 0\}$, $\{y \in \mathcal{Y}^+, x \in \mathcal{X}_y^+\}$.

Now we find the equality case. We require $\sum_{y \in \mathcal{Y}_x^+} p(y) = \sum_{y \in \mathcal{Y}} p(y)$ for all $x \in \mathcal{X}^+$ from the last inequality (the previous case is required, since otherwise the argument in the log is zero, and we cannot and $p(x, y)/p(y)$ is constant over $y \in \mathcal{Y}_x^+$ for all $x \in \mathcal{X}^+$. If $a, b, c, d \in \mathbb{R}^\times$ with $a/b = c/d = k$, then $\frac{a+c}{b+d} = k$. Using the reasonable generalization, we have for fixed x, y that

$$\frac{p(x, y)}{p(y)} = \frac{\sum_{y \in \mathcal{Y}_x^+} p(x, y)}{\sum_{y \in \mathcal{Y}_x^+} p(y)} = \frac{\sum_{y \in \mathcal{Y}_x^+} p(x, y)}{\sum_{y \in \mathcal{Y}} p(y)} = \sum_{y \in \mathcal{Y}_x^+} p(y|x) p(x) = p(x), \quad (20)$$

so $p(x, y) = p(x)p(y)$ for all $x \in \mathcal{X}^+, y \in \mathcal{Y}_x^+$, ie X, Y are independent. Conversely, we can see that if X, Y are independent, then $H(X|Y) = H(X)$. $\square$

Corollary 2.6. $H(X, Y) \leq H(X) + H(Y)$

In other words, the uncertainty of (X, Y) is less than the sum of the uncertainties of the marginals. Next, we have the following proposition, which is related to the Data Processing inequality, which we introduce later.

Proposition 2.7. *Let f be a function. Then $H(fX) \leq H(X)$ with equality iff f is injective on $\mathcal{X}^+$.*

Proof. Note the absence of constraints on f . The intuition is that f can sometimes group values of X together, decreasing entropy. Otherwise, f moves around values of X and doesn't affect entropy, since entropy depends on the probabilities (not values).

Denote by $\text{cod } f$ the codomain of f . We have:

$$-H(fX) = \sum_{y \in (\text{cod } f)^+} p(y) \log p(y) = \sum_{y \in (\text{cod } f)^+} \left(\sum_{x \in f^{-1}y} p(x) \log \sum_{x \in f^{-1}y} p(x) \right) \quad (21)$$

$$\geq \sum_{y \in (\text{cod } f)^+} \sum_{x \in f^{-1}y} p(x) \log p(x) = \sum_{x \in \mathcal{X}^+} p(x) \log p(x) = -H(X), \quad (22)$$

as desired. We applied the more general version of the following inequality: for $a, b \in \mathbb{R}^+$, $(a+b) \log(a+b) = a \log(a+b) + b \log(a+b) > a \log a + b \log b$. If f is not injective on $\mathcal{X}^+$, then the inequality in the second line will be strict (the one-dimensional form of this inequality is $a \log a = a \log a$).

Here is the sketch of another solution. The map $x \mapsto (x, f(x))$ preserves all information of X since the map is bijective. Thus,

$$H(X) = H(X, f(X)) = H(f(X)) + H(X|f(X)) \geq H(f(X)) \quad (23)$$

so it remains to find when $H(X|f(X)) = 0$. This is left to the reader. $\square$

Finally, we introduce the chain rule for entropy.

Proposition 2.8. *Let $X_1, X_2, \dots, X_n$ be r.v.s. Then $H(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i|X_{<i})$, where $H(X_1|X_{<1}) = H(X_1)$.*

Proof. In other words, the uncertainty of $(X_1, X_2, \dots, X_n)$ equals the uncertainty of X_1 added to the uncertainty of X_2 reduced by X_1 , and so on. $n = 2$ specializes to Proposition 2.2

The proof is by induction. $n = 1$ holds. Suppose the result for n . Then

$$H(X_1, X_2, \dots, X_{n+1}) = H(X_1, X_2, \dots, X_n) + H(X_{n+1}|X_1, X_2, \dots, X_n) \quad (24)$$

$$= \sum_{i=1}^n H(X_i|X_{<i}) + H(X_{n+1}|X_{<n+1}) = \sum_{i=1}^{n+1} H(X_i|X_{<i}), \quad (25)$$

as desired. $\square$

3 Mutual Information

Suppose we have two r.v.s X, Y . The mutual information $I(X; Y)$ is a number that captures the degree to which information about X applies to Y . As the name suggests, this quantity is

symmetric with respect to (wrt) X, Y . Intuitively, if $X = Y$, then $I(X; Y)$ should be relatively large. We define $I(X; Y) = H(X) - H(X|Y)$. We know $H(X|Y) \leq H(X)$ by Proposition 2.5, so $I(X; Y) \geq 0$. Equality holds iff $H(X|Y) = H(X)$, ie X, Y are independent. Intuitively, $I(X; Y)$ measures the decrease in uncertainty of X given Y .

Proposition 3.1. $I(X; Y) = H(X) + H(Y) - H(X, Y)$.

Proof. Apply Proposition 2.2. As a corollary, $I(X; Y)$ is symmetric. $\square$

Proposition 3.2. $I(X, Y) = \sum_{x \in \mathcal{X}, y \in \mathcal{Y} p(x, y) \neq 0} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$.

Proof. This can be proven by expanding $H(X) + H(Y) - H(X, Y)$ into sums. $\square$

Proposition 3.3. $I(X; Y) \leq \min(H(X), H(Y))$.

Proof. By Proposition 2.1, we have $H(X_y) \geq 0$, so $H(X|Y) = \sum_{y \in \mathcal{Y}^+} H(X_y)p(y) \geq 0$, so $I(X; Y) \leq H(X)$. The other part is similar. $\square$

Proposition 3.4. $I(X; X) = H(X)$.

We define conditional mutual information:

$$I(X; Z|Y) = \sum_{y \in \mathcal{Y}^+} I(X; Z|Y = y)p(y) \quad (26)$$

$$= \sum_{\substack{(x, y, z) \in \mathcal{X} \times \mathcal{Y} \times \mathcal{Z} \\ p(x, y, z) \neq 0}} p(x, y, z) \log \frac{p(x, z|y)}{p(x|y)p(z|y)} \quad (27)$$

Note the names of the variables are not “in order.” The reasons will be apparent later.

Proposition 3.5. $I(X; Z|Y) \geq 0$ with equality iff X, Z are conditionally independent given Y . X, Z are conditionally independent given Y , denoted by $X \perp Z|Y$ if $p(x, z|y) = p(x|y)p(z|y)$ whenever $p(x, y, z) \neq 0$.

Proof. The inequality is clear. Suppose equality holds. We have $0 = I(X; Z|Y) = \sum_{y \in \mathcal{Y}^+} I(X; Z|Y = y)p(y)$, so by Proposition 2.5, we have that $X_y \perp Z_y$ for all $y \in \mathcal{Y}^+$, ie $X \perp Z|Y$. $\square$

Proposition 3.6. $I(X; Y, Z) = I(X; Y) + I(X; Z|Y)$.

Proof. From now on, we will be more informal with the summing, using expectations to handle them. We have

$$I(X; Y, Z) = \mathbb{E}_{(x, y, z) \sim p} \log \frac{p(x, y, z)}{p(x)p(y, z)} \quad (28)$$

$$I(X; Y) = \mathbb{E}_{(x, y, z) \sim p} \log \frac{p(x, y)}{p(x)p(y)} \quad (29)$$

$$I(X; Z|Y) = \mathbb{E}_{(x, y, z) \sim p} \log \frac{p(x, z|y)}{p(x|y)p(z|y)} \quad (30)$$

so by Linearity of Expectation, it suffices to show

$$\frac{p(x, y, z)}{p(x)p(y, z)} = \frac{p(x, y)}{p(x)p(y)} \cdot \frac{p(x, z|y)}{p(x|y)p(z|y)} \quad (31)$$

when $p(x, y, z) \neq 0$. Indeed, the right hand side (RHS) expands as

$$\frac{p(x, y)}{p(x)p(y)} \cdot \frac{p(x, y, z)p(y)}{p(x, y)p(y, z)} = \frac{p(x, y, z)}{p(x)p(y, z)}. \quad (32)$$

The proposition can be interpreted as follows: the information between X and (Y, Z) is the sum of the information between X, Y and that of X, Z when Y is known. $\square$

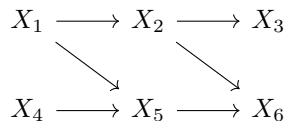
4 Data-Processing Inequality

The Data-Processing inequality shows that we cannot modify r.v.s to gain more information from them. In some sense it is similar to Proposition 2.7. Suppose we have r.v.s X, Y, Z . We will say the sequence (chain) $X \rightarrow Y \rightarrow Z$ is Markov if $X \perp Z|Y$, ie $p(x|y)p(z|y) = p(x, z|y)$ for all $y \in \mathcal{Y}^+$ such that $p(x, z|y) \neq 0$. This is equivalent to requiring

$$p(x, y)p(z, y) = p(x, y, z)p(y) \quad (33)$$

if $p(x, y, z) \neq 0$. This condition is symmetric with respect to X, Z so $X \rightarrow Y \rightarrow Z$ is the same as $Z \rightarrow Y \rightarrow X$. In some sense, if we ask about Z , we only need to remember the previous state Y ; we don't need to know about X . This is called the *memoryless* property of Markov chains.

The Markov chain $X \rightarrow Y \rightarrow Z$ is a special case of Bayesian networks. This isn't directly related to the topic, but is an important application in machine learning. Let G be a directed acyclic graph (DAG) with nodes $\{X_i\}_{i \in [n]}$, where the X_i are r.v.s. For a node label $i \in [n]$ denote by Pa_i the parents (also represented by labels, as a subset of $[n]$). For example, if we have G :



then $\text{Pa}_1 = \emptyset, \text{Pa}_5 = \{1, 4\}$. Note that there need not be a single node with no parent, nor a single node with no children. G is said to be a Bayesian network if the joint distribution of the X_i factors as:

$$p(x_1, x_2, \dots, x_n) = \prod_{i \in [n]} p(x_i | x_{\text{Pa}_i}), \quad (34)$$

where x_{Pa_i} denotes that the conditionals are taken over the parents of node i . For the above network, we have

$$p(x_1, x_2, \dots, x_6) = p(x_1)p(x_2|x_1)p(x_3|x_2)p(x_4)p(x_5|x_1, x_4)p(x_6|x_2, x_5) \quad (35)$$

Now we return to the Data-processing inequality.

Proposition 4.1. *Let g be a function and $Z = f(Y)$. Then $X \rightarrow Y \rightarrow Z$ is Markov.*

Proof. Suppose $z = f(y)$. Then both sides of Equation 33 become $p(x, y)p(y)$. If $z \neq f(y)$, then both sides become 0. $\square$

Theorem 4.2 (Data-processing inequality). $I(X, Y) \geq I(X, Z)$.

Proof. Consider the quantity $I(X; Y, Z)$, symmetric with respect to Y, Z . We expand in two ways using Proposition 3.6:

$$I(X; Y, Z) = I(X; Y) + I(X; Z|Y) \tag{36}$$

$$= I(X; Z) + I(X; Y|Z) \tag{37}$$

We know by that Markovity is equivalent to $I(X; Z|Y) = 0$ by Proposition 3.5. Another application of 3.5 gives $I(X; Y|Z) \geq 0$, so the inequality follows. We can think that if we have some fixed quantity $C = A + B$, if we decrease A , then we must increase B , though we can also explicitly write out the inequalities. $\square$

Theorem 4.3 (Shearer's inequality).

References

- [CT06] Thomas M. Cover and Joy A. Thomas, *Elements of information theory*, 2nd ed., Wiley-Interscience, Hoboken, NJ, 2006.
- [Pol16] Yury Polyanskiy, *6.441 information theory, spring 2016: Problem set 1*, <https://ocw.mit.edu/courses/6-441-information-theory-spring-2016/pages/assignments/>, 2016, MIT OpenCourseWare, Massachusetts Institute of Technology.